

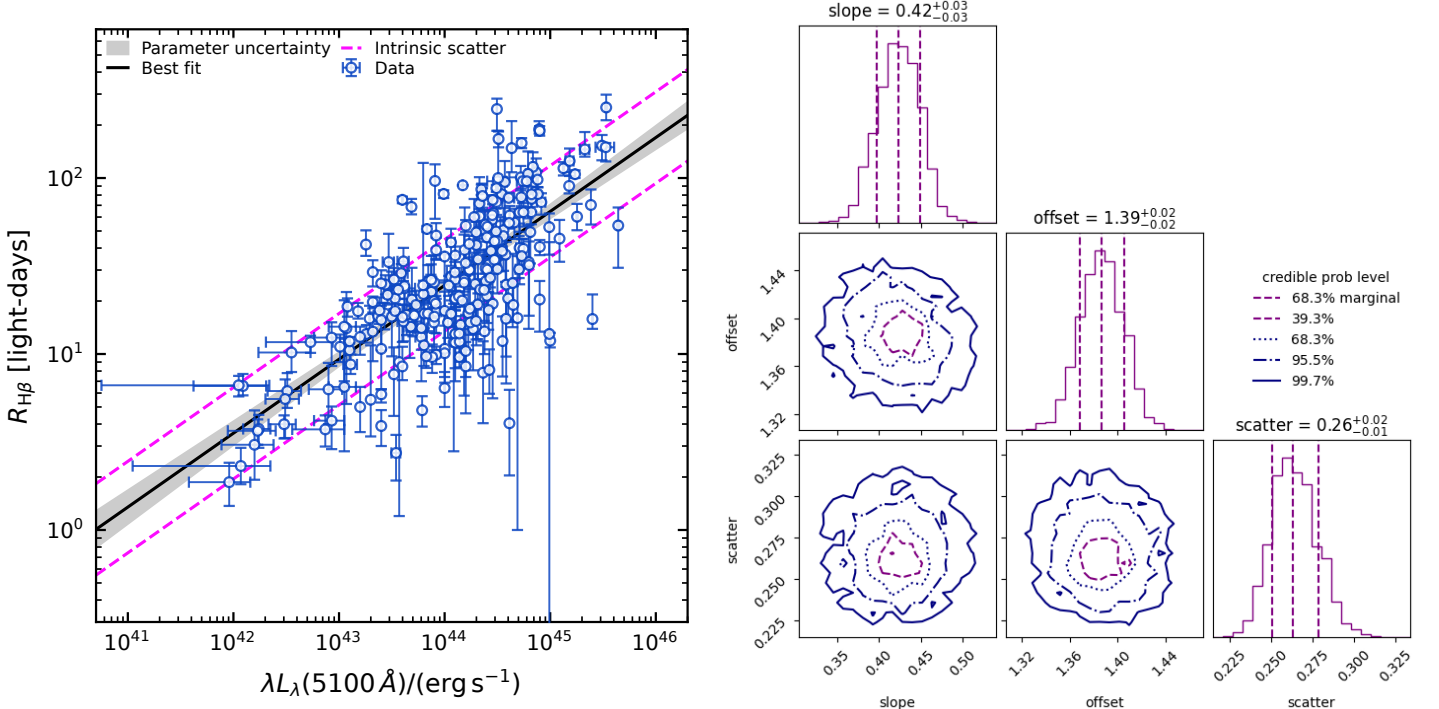


Figure 1: Best-fit of the H β R - L relation for the whole sample included in the database. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

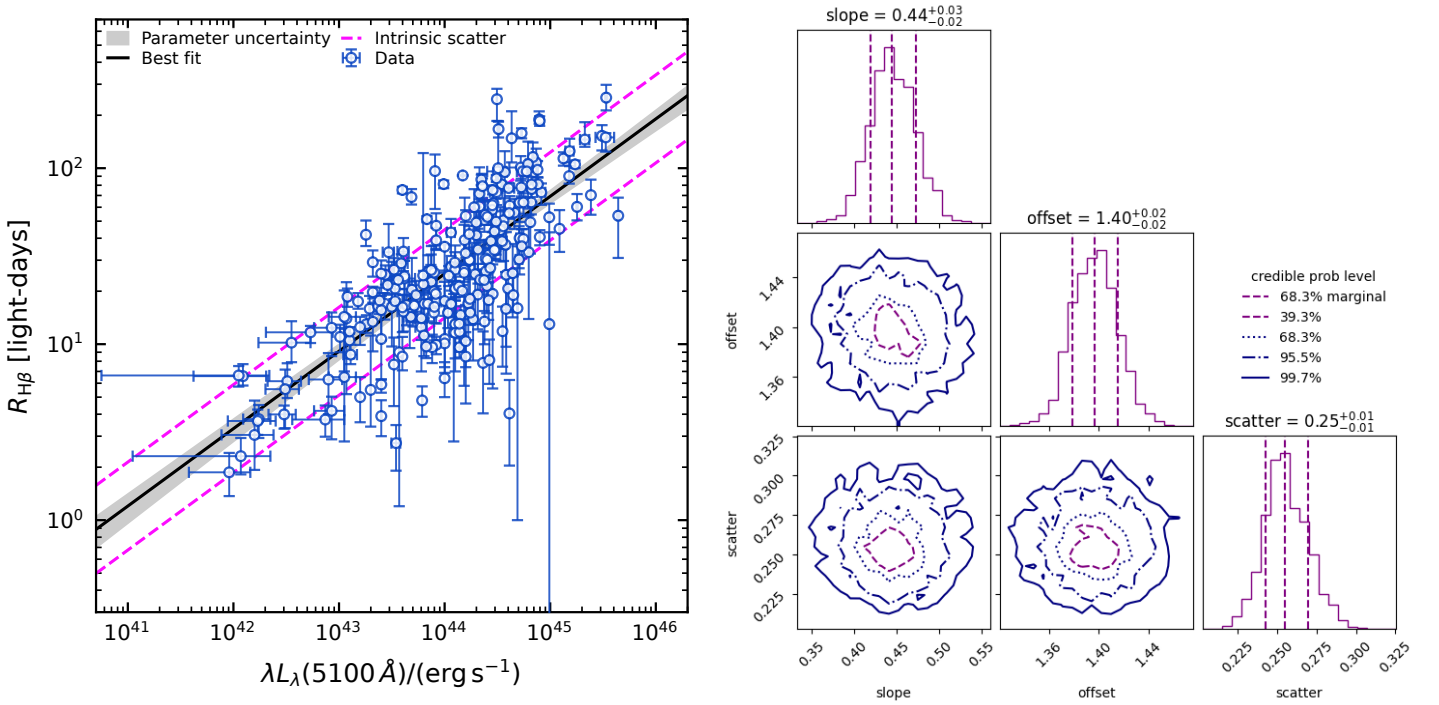


Figure 2: Best-fit of the H β R - L relation for the sample consisting of Cat. 0+1+2 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

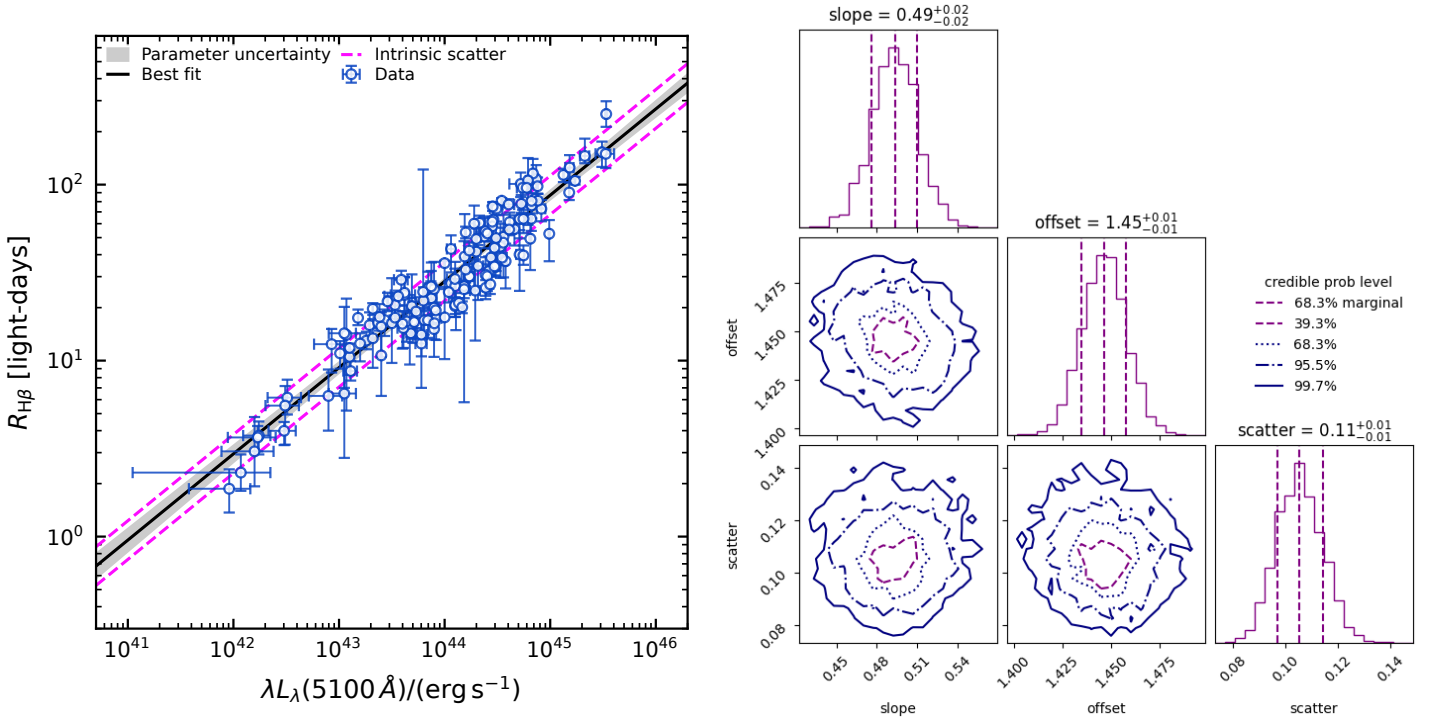


Figure 3: Best-fit of the H β R - L relation for the sample consisting of Cat. 0+1 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

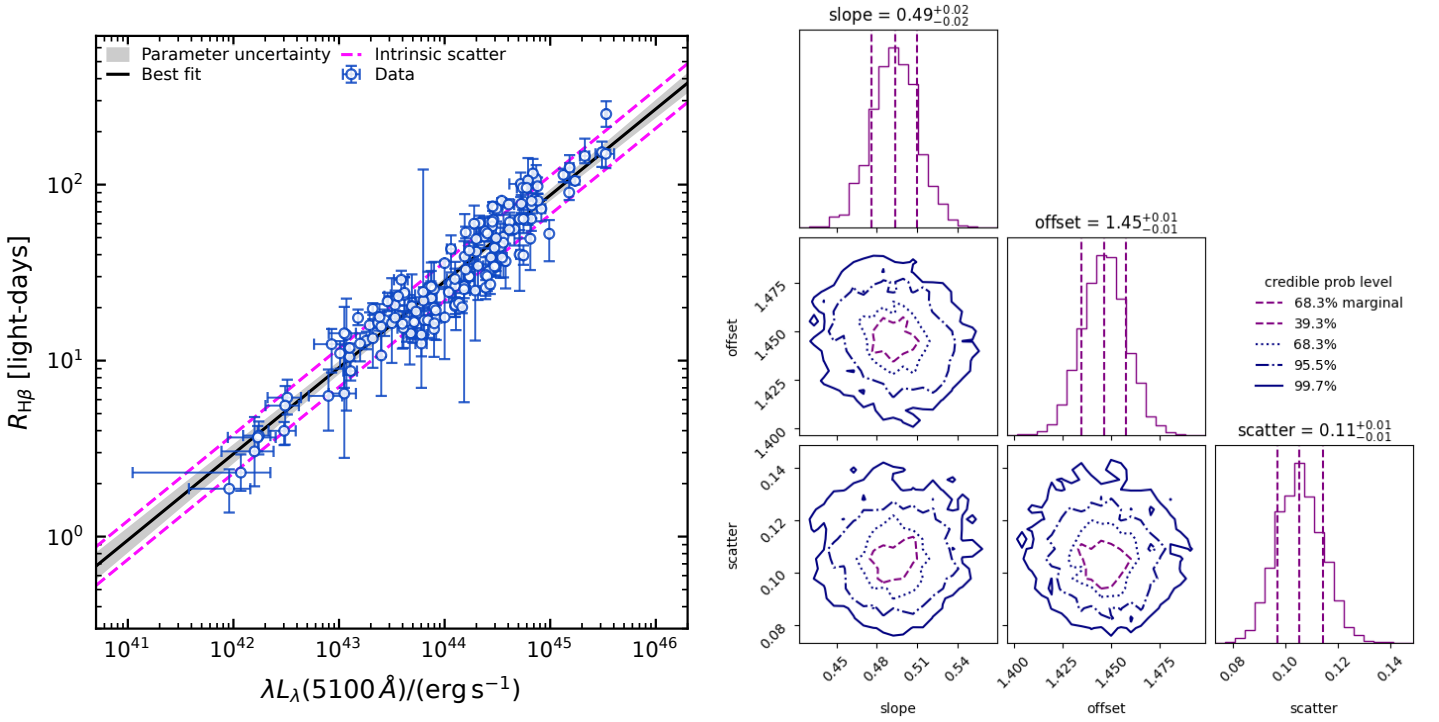


Figure 4: Best-fit of the $H\beta$ $R-L$ relation for the sample consisting of Cat. 0 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

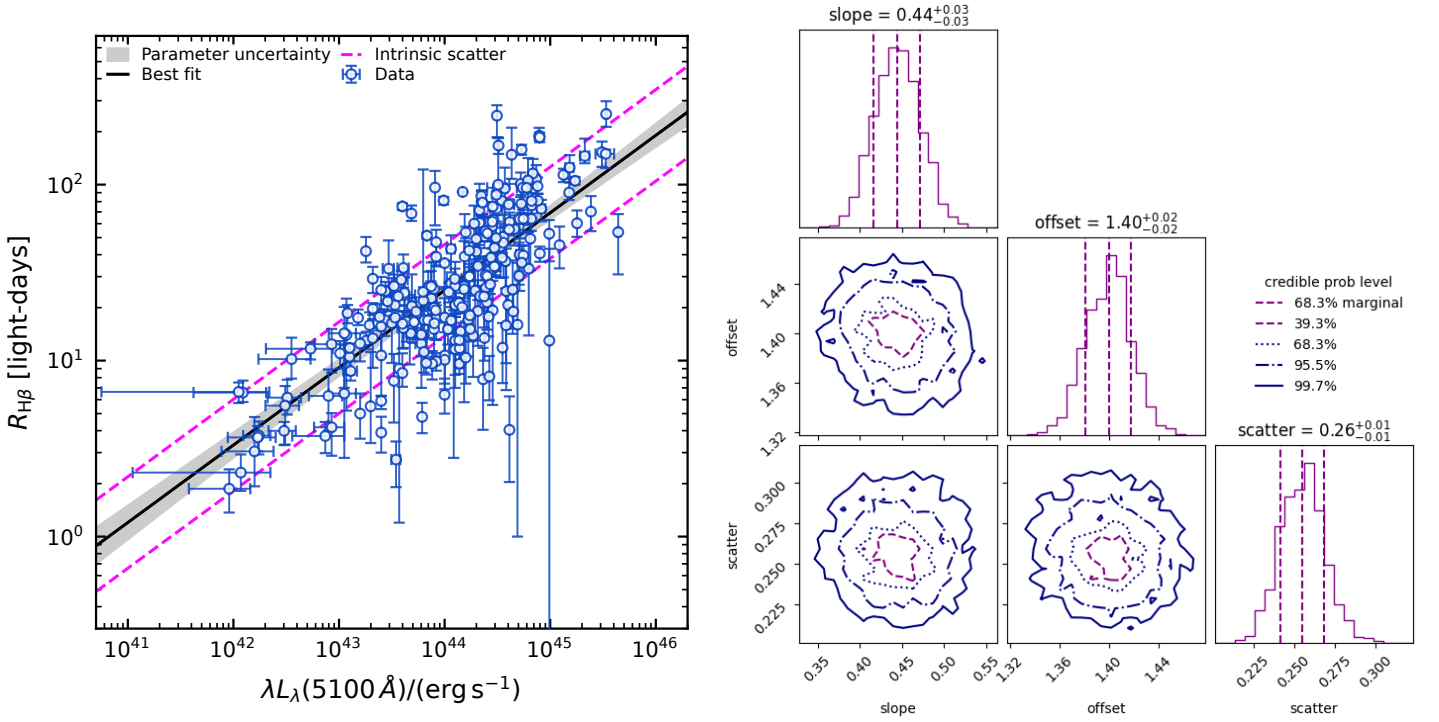


Figure 5: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0+1+2 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.7/1.3. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

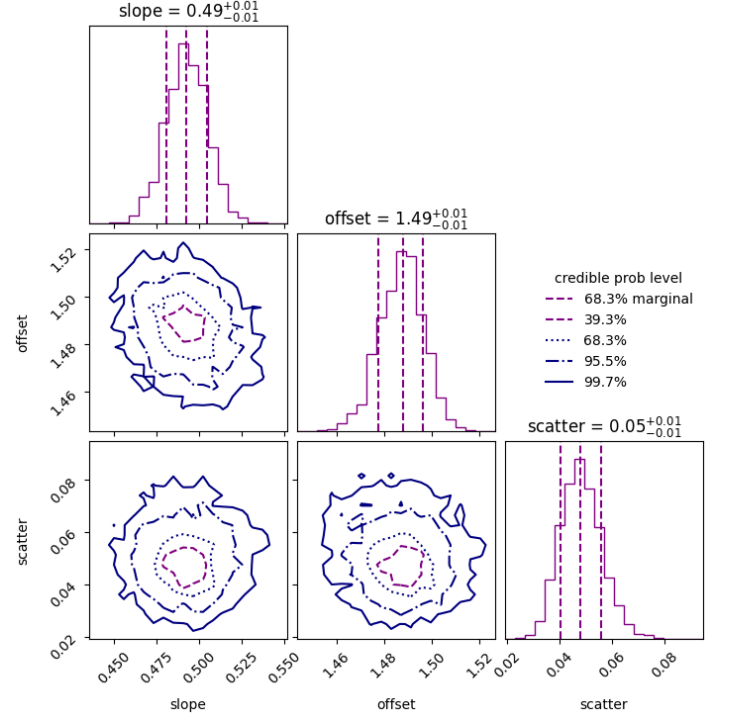
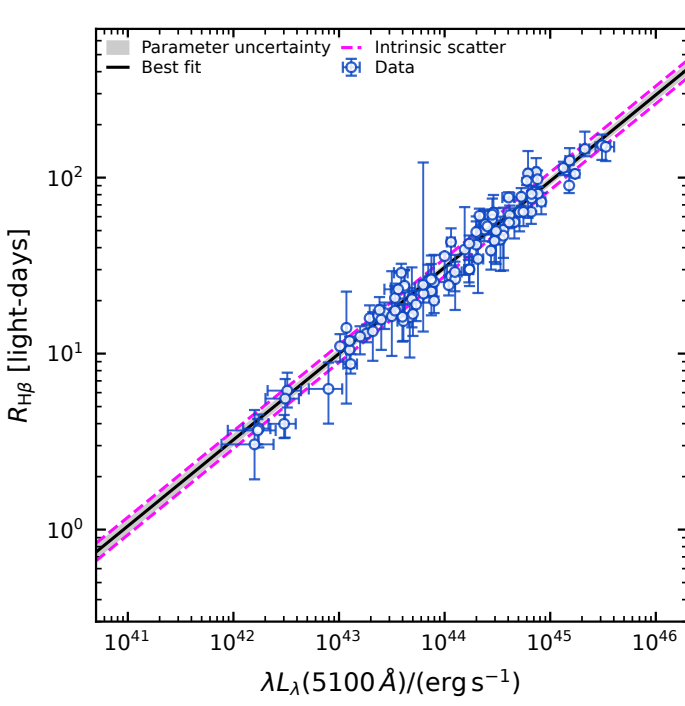


Figure 6: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0+1 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.7/1.3. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

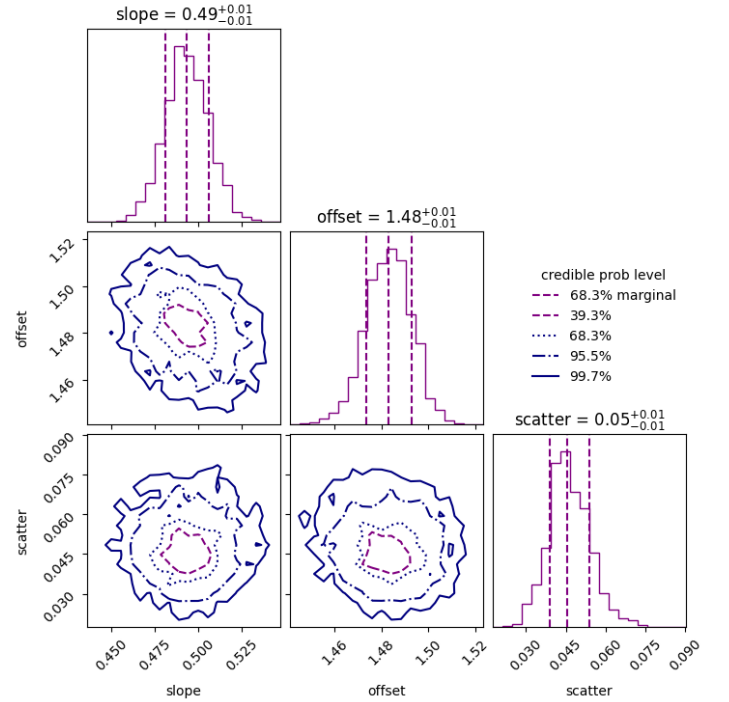
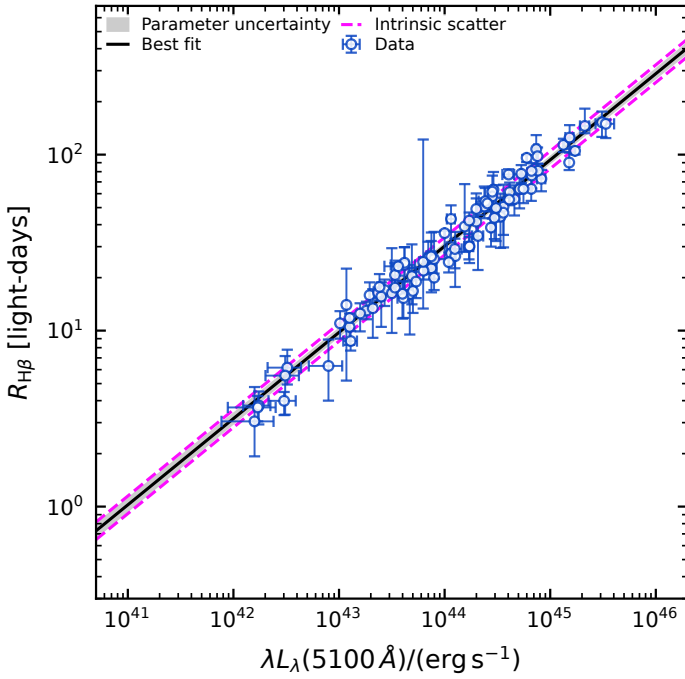


Figure 7: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.7/1.3. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

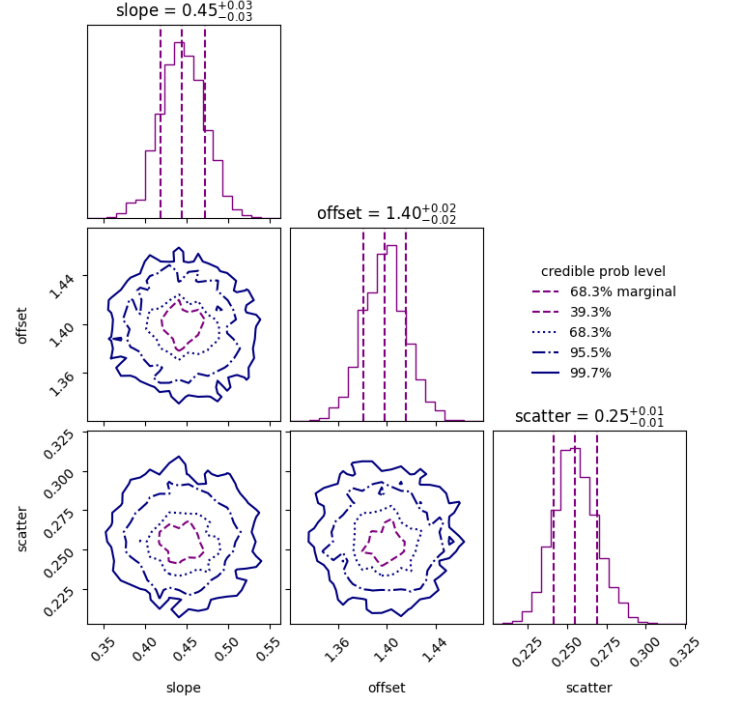
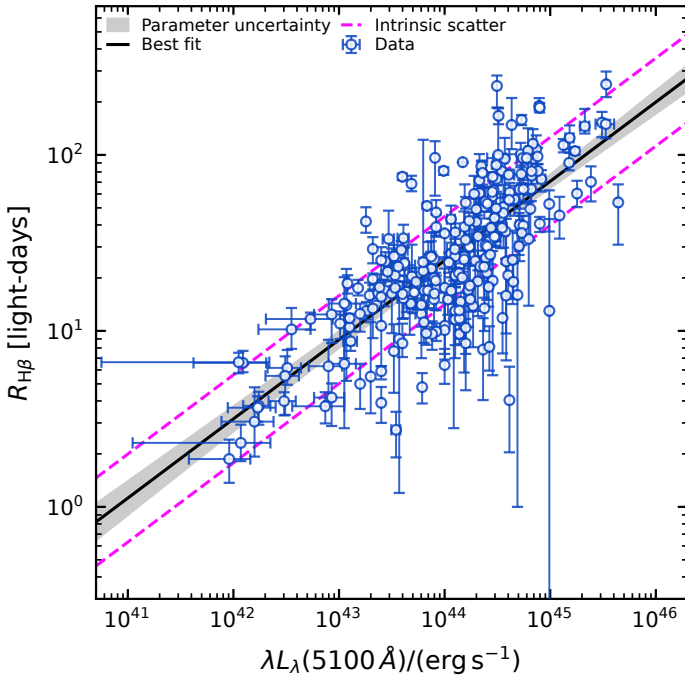


Figure 8: Best-fit of the H β R - L relation for the sample consisting of Cat. 0+1+2 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.3/1.7. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

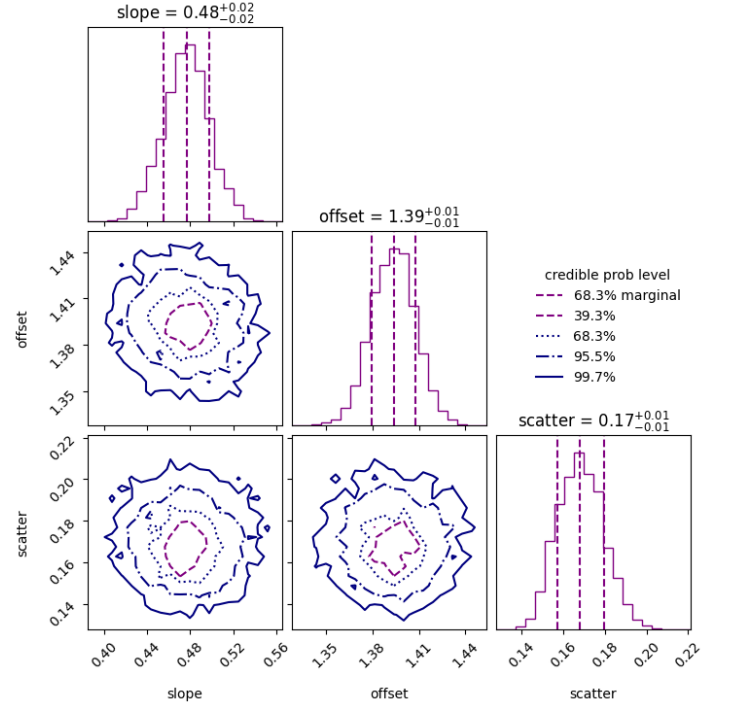
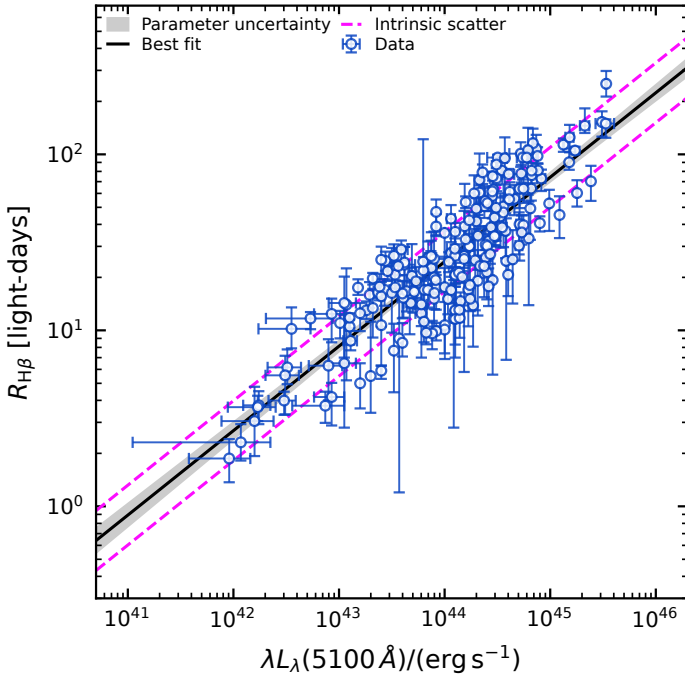


Figure 9: Best-fit of the H β R - L relation for the sample consisting of Cat. 0+1 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.3/1.7. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

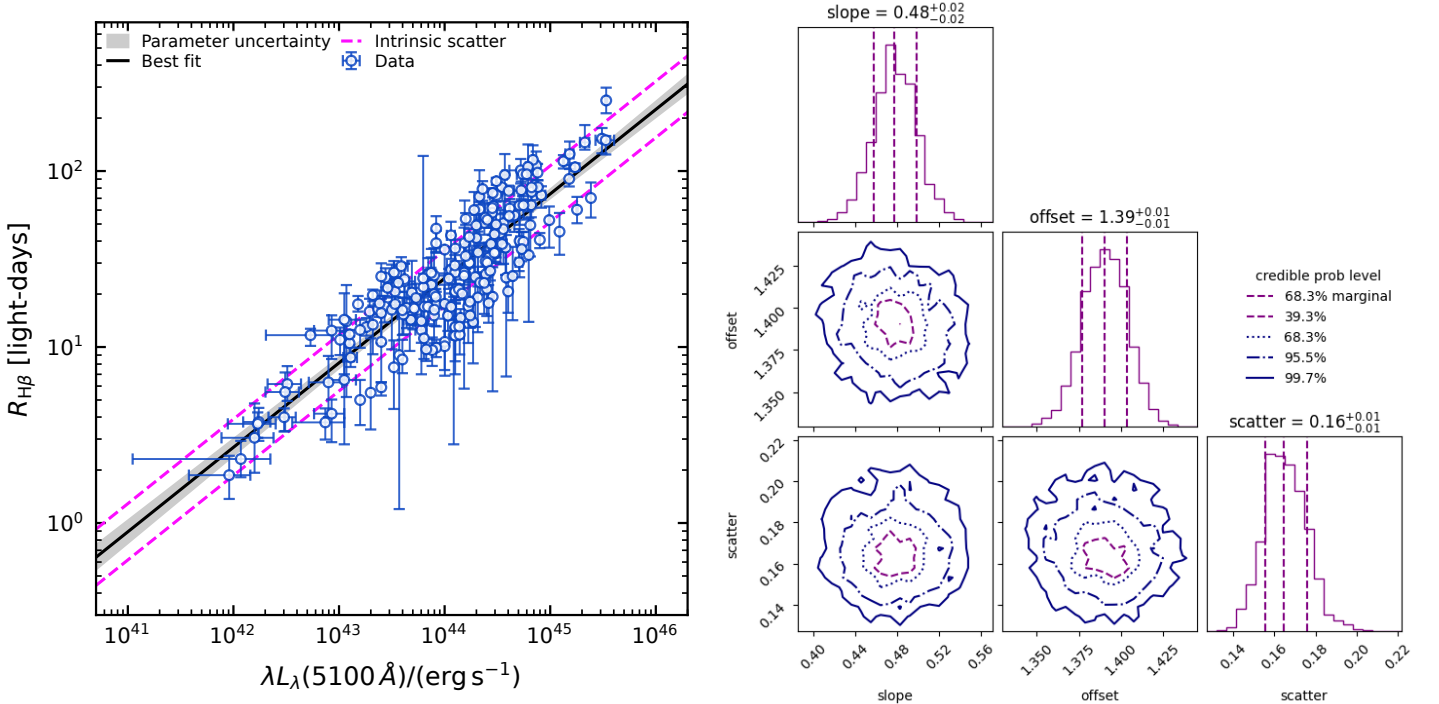


Figure 10: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0 assuming $\alpha_{\text{ref}} = 0.5$ and thresholds 0.3/1.7. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

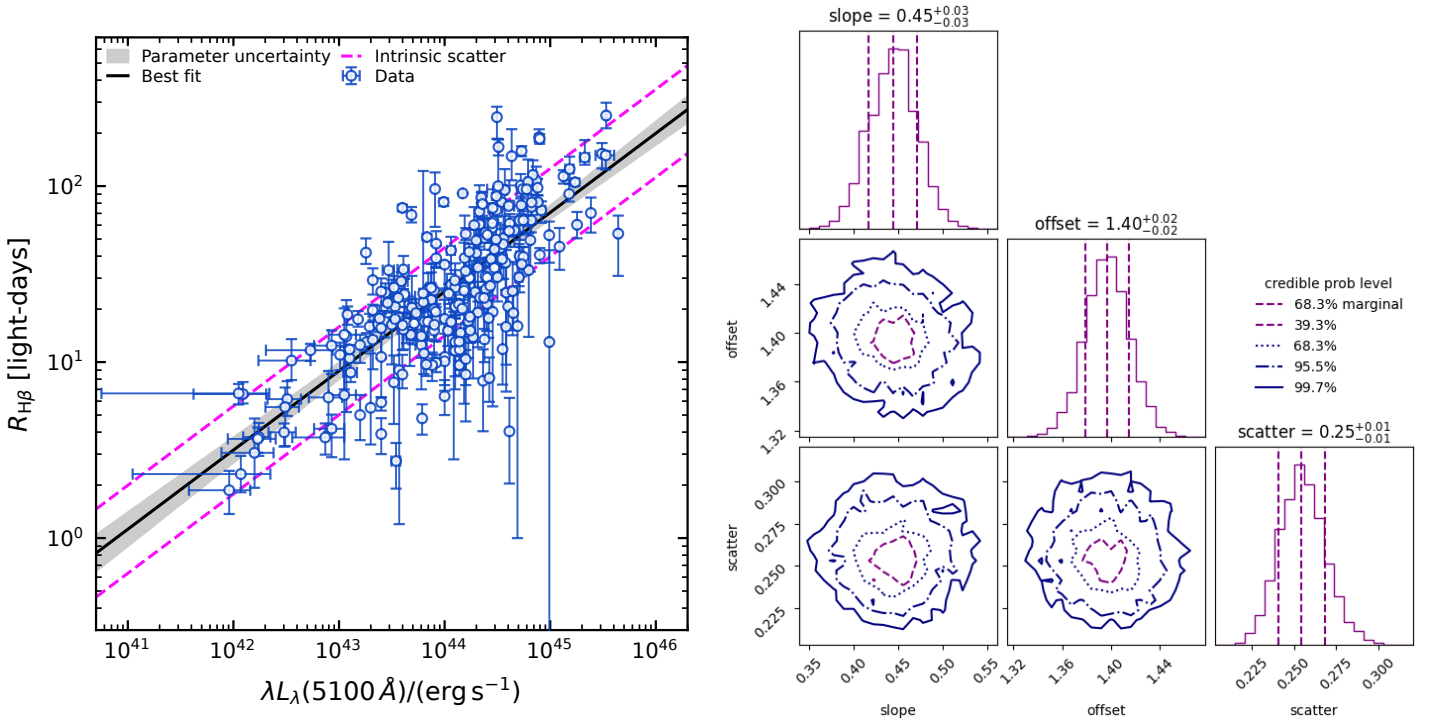


Figure 11: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0+1+2 assuming $\alpha_{\text{ref}} = \frac{1}{3}$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

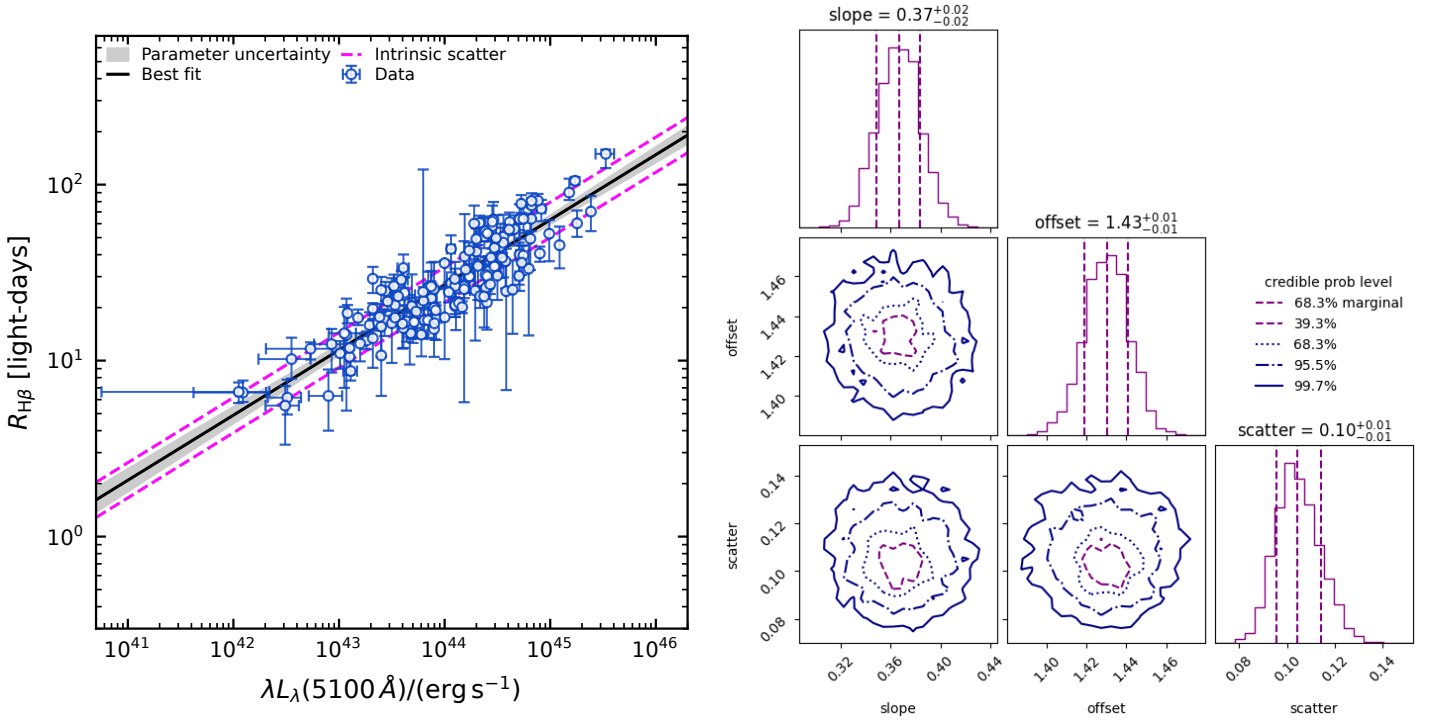


Figure 12: Best-fit of the H β R - L relation for the sample consisting of Cat. 0+1 assuming $\alpha_{\text{ref}} = \frac{1}{3}$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

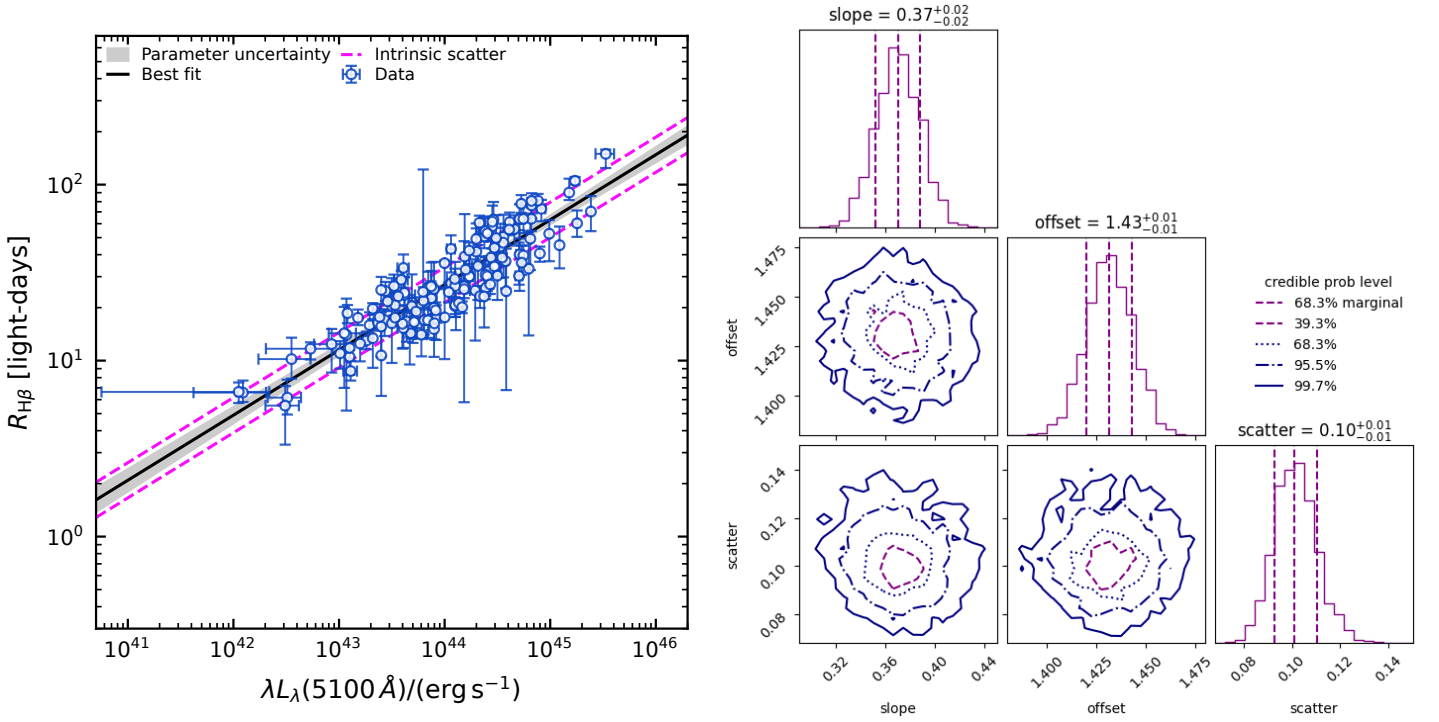


Figure 13: Best-fit of the H β R - L relation for the sample consisting of Cat. 0 assuming $\alpha_{\text{ref}} = \frac{1}{3}$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

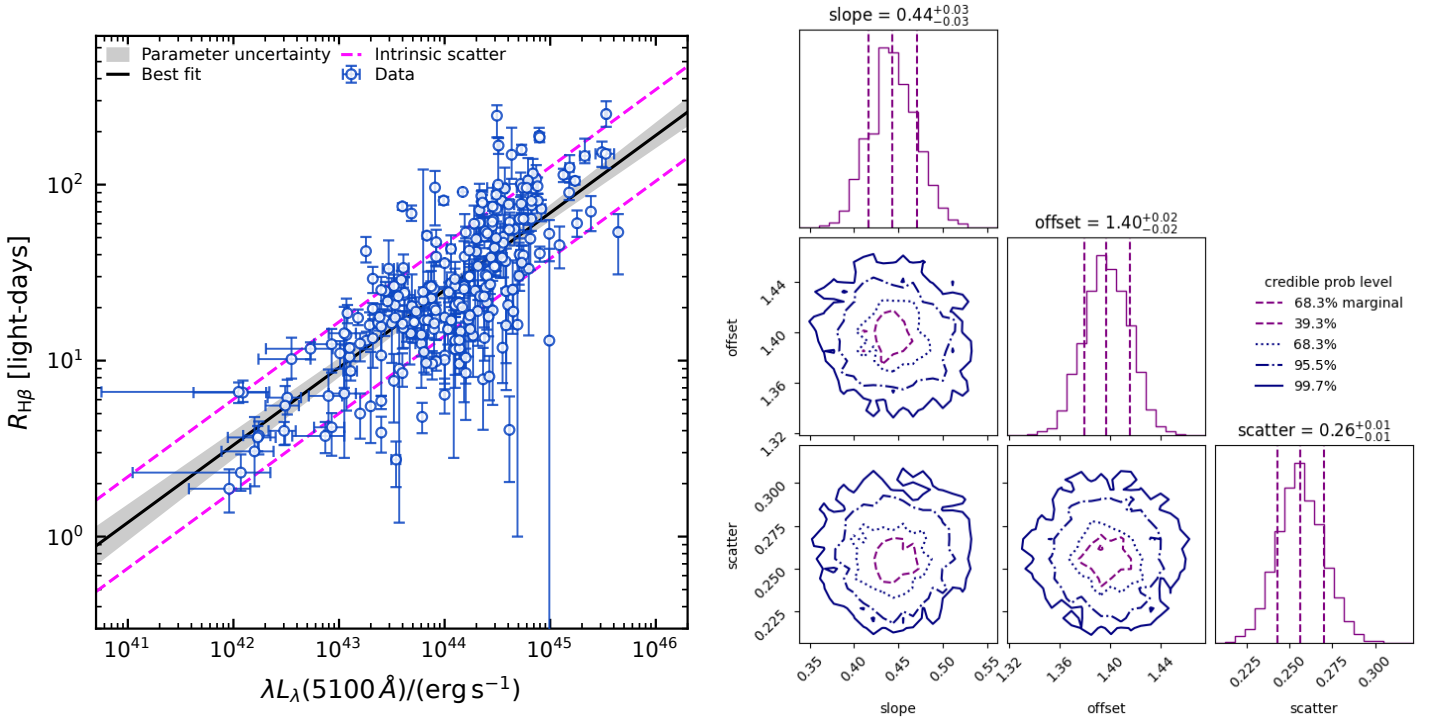


Figure 14: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0+1+2 assuming $\alpha_{\text{ref}} = 0.7$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

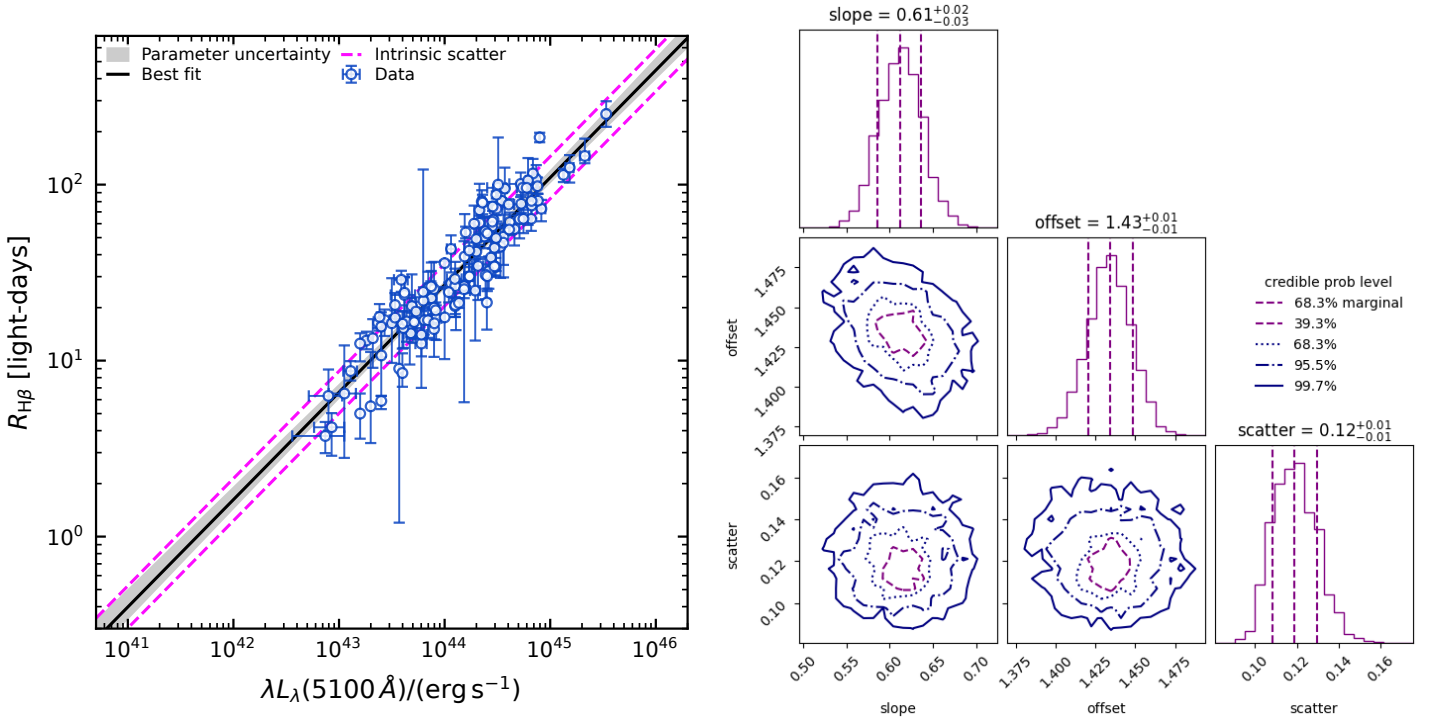


Figure 15: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0+1 assuming $\alpha_{\text{ref}} = 0.7$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

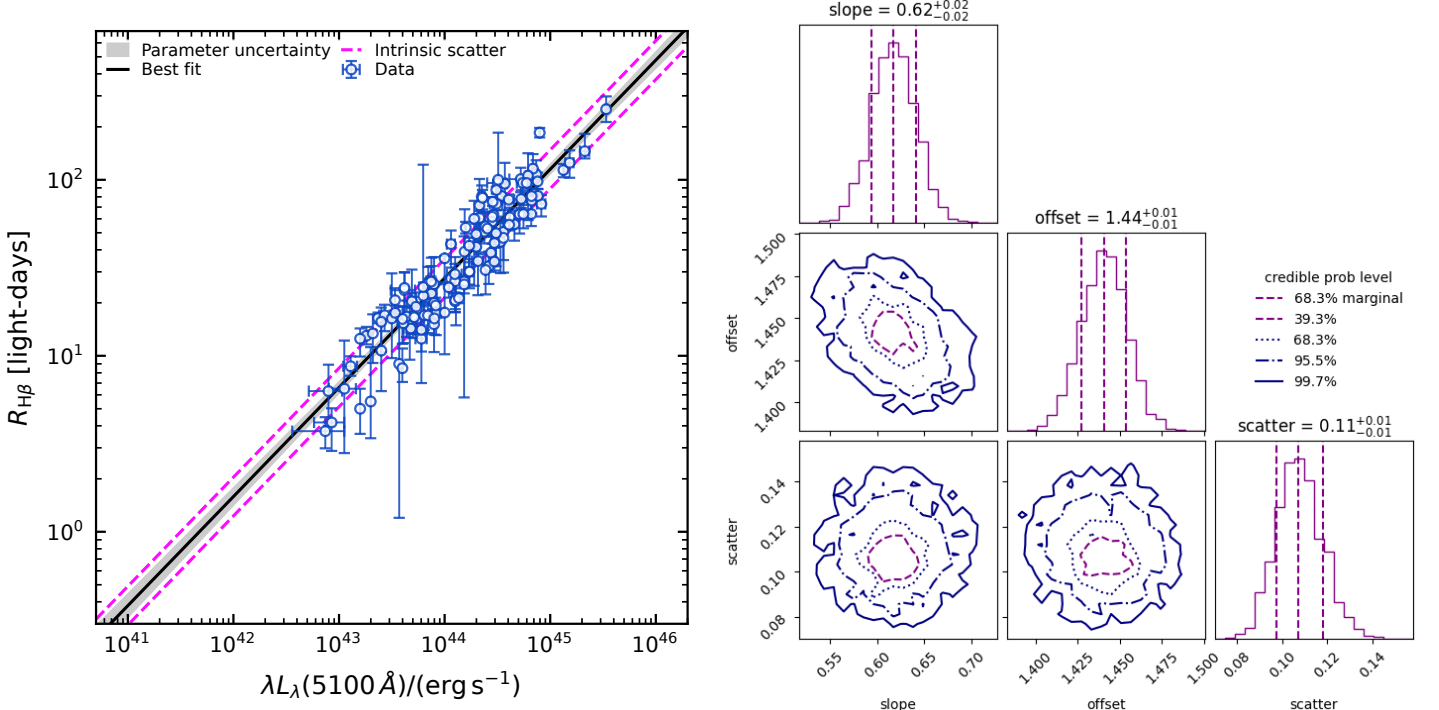


Figure 16: Best-fit of the $H\beta$ R - L relation for the sample consisting of Cat. 0 assuming $\alpha_{\text{ref}} = 0.7$ and thresholds 0.5/1.5. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

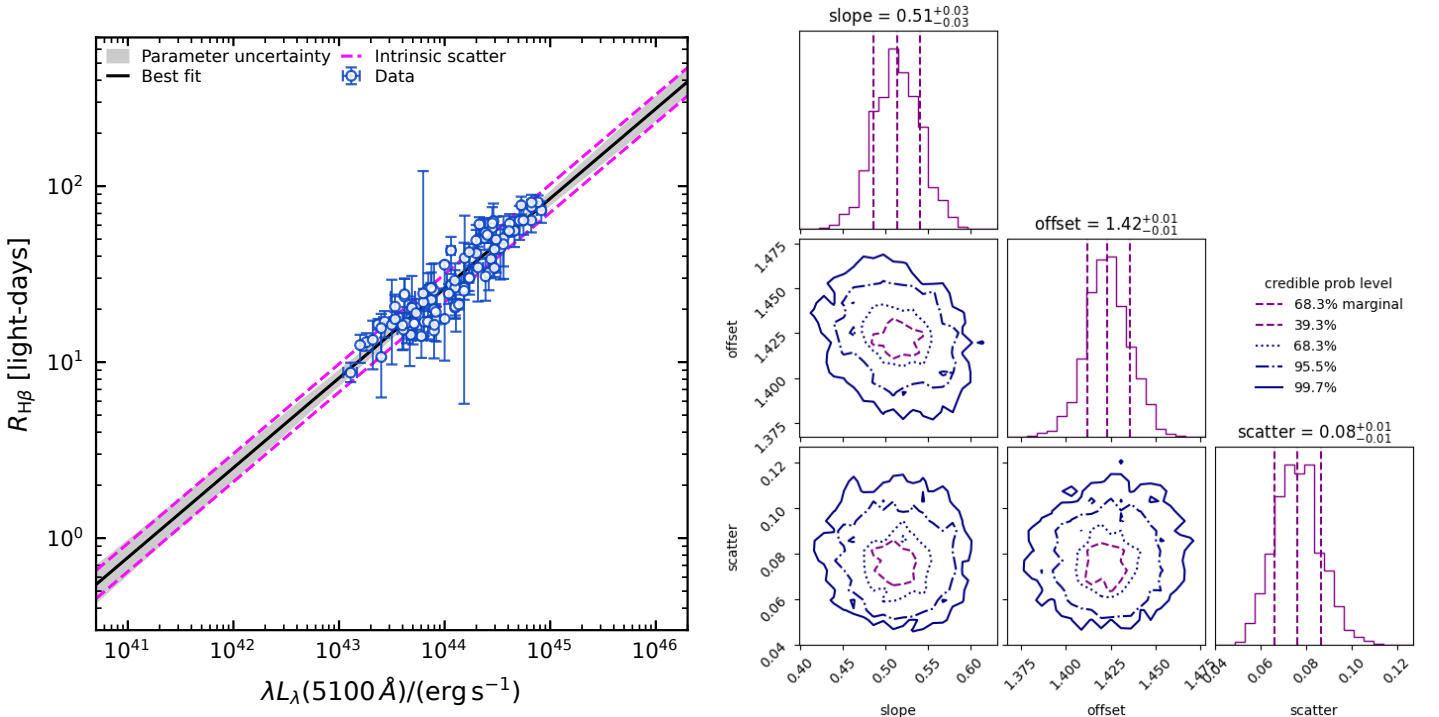


Figure 17: Best-fit of the $H\beta$ R - L relation for consensus sample 1. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

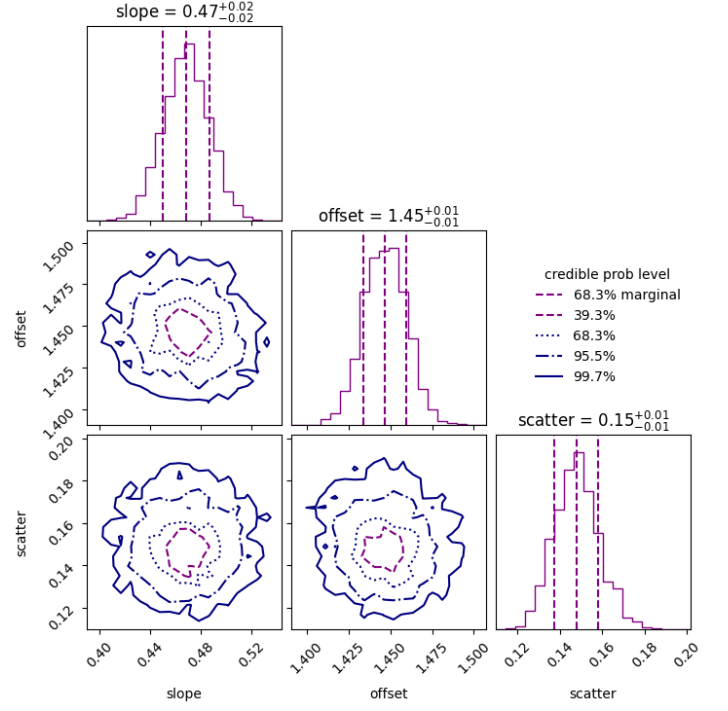
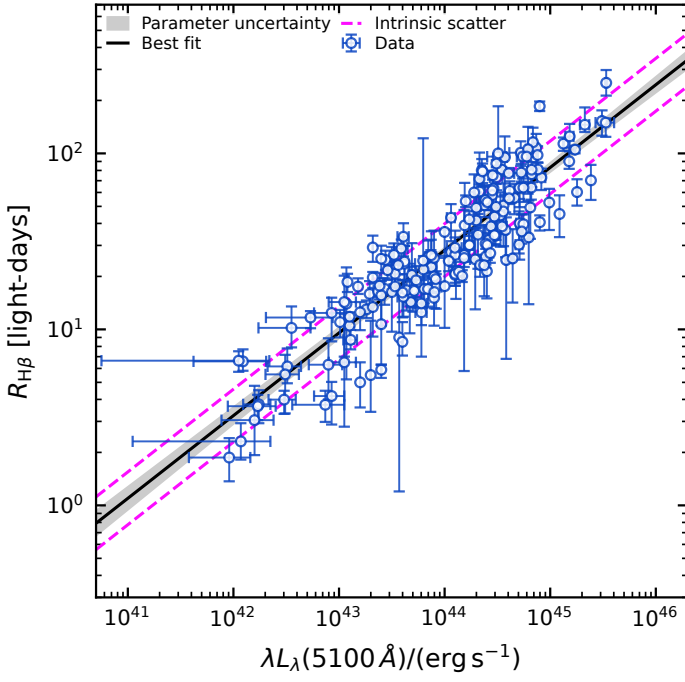


Figure 18: Best-fit of the $H\beta$ R - L relation for consensus sample 2. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

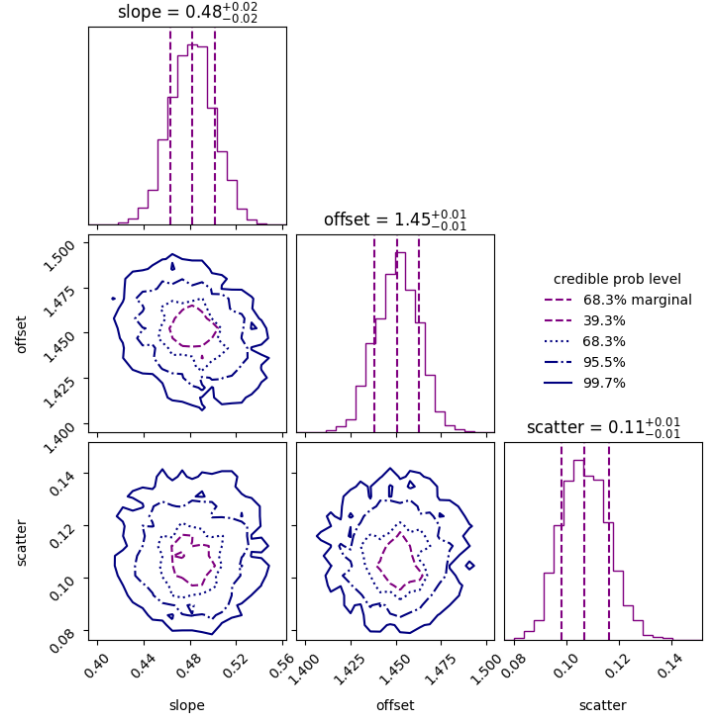
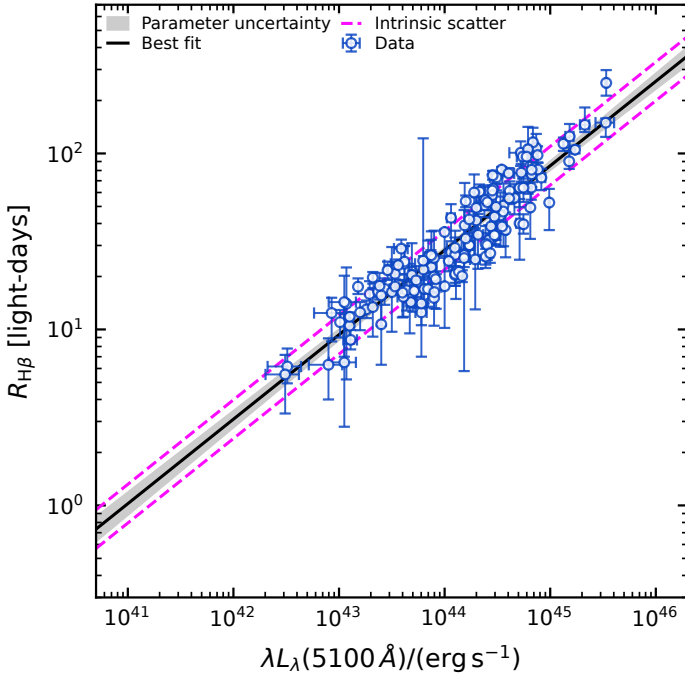


Figure 19: Best-fit of the $H\beta$ R - L relation for consensus sample 3. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

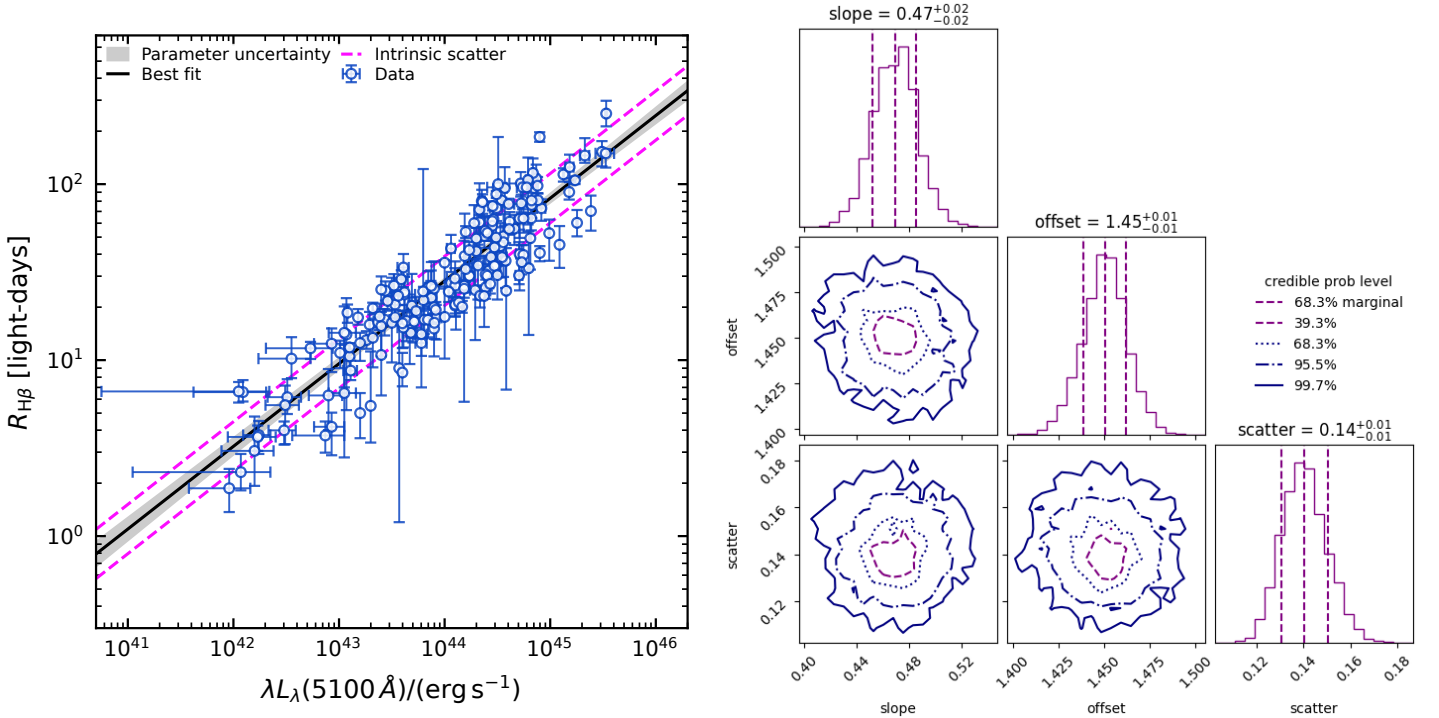


Figure 20: Best-fit of the $H\beta$ R - L relation for consensus sample 4. The black solid lines shows the best linear fit, gray shaded areas indicate the uncertainty of the fit, magenta dashed lines show fitted intrinsic scatter.

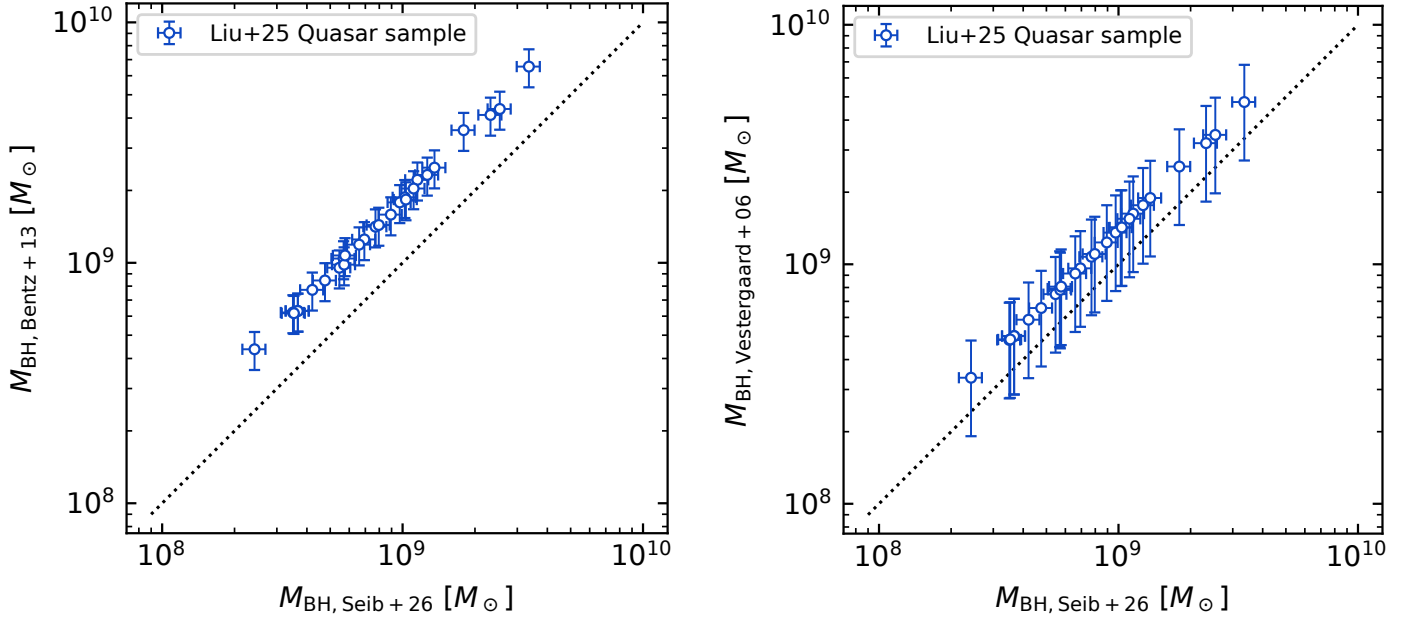


Figure 21: Comparison of black hole masses for the high- z quasar sample of Liu et al. (2025), inferred with literature SE prescriptions and with this work's estimator. From left to right the panels are: (1) SE mass estimator using Bentz et al. (2013) calibrated R - L relation; (2) SE mass estimator fitted by Vestergaard & Peterson (2006). Uncertainties reflect the intrinsic scatter of each calibration, which dominates over parameter and measurement uncertainties.

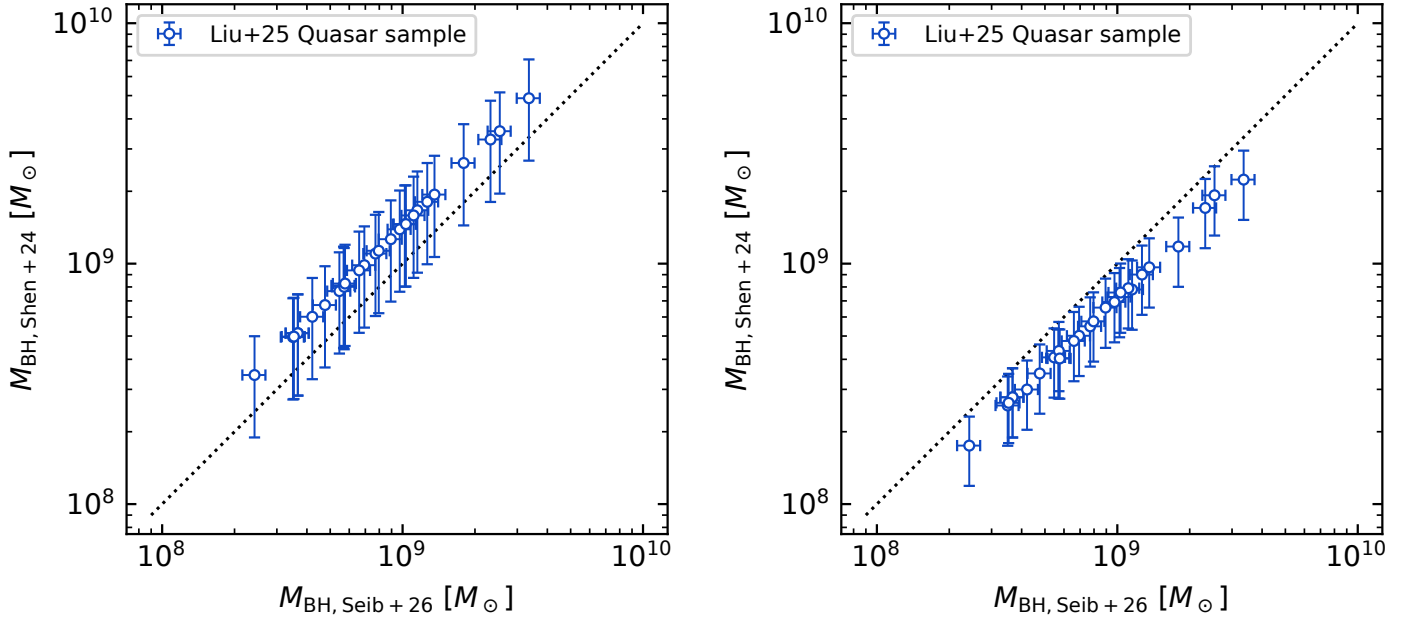


Figure 22: Comparison of black hole masses for the high- z quasar sample of Liu et al. (2025), inferred with literature SE prescriptions and with this work’s estimator. From left to right the panels are: (1) SE mass estimator fitted by Shen et al. (2024); (2) SE mass estimator calculated from Shen et al. (2024) calibrated R – L relation. Uncertainties reflect the intrinsic scatter of each calibration, which dominates over parameter and measurement uncertainties.